

question 1

```
int ECSumList(const vector<int> &listNums)
{
    // your code goes here ...
    int sum = 0;
    for (size_t i = 0; i < listNums.size(); i++) {
        sum += listNums[i];
    }
    return sum;
}
```

```
karin@Karinas_Laptop CLANG64
$ clang++ ECSumListVec.cpp
```

```
karin@Karinas_Laptop CLANG64
$ ./a.exe
sum: 9
```

```
int ECSumList(const vector<int> &listNums)
{
    // your code goes here ...
    int sum = 0;
    for (int num : listNums) {
        sum += num;
    }
    return sum;
}
```

```
karin@Karinas_Laptop CLANG64
$ clang++ ECSumListVec.cpp
```

```
karin@Karinas_Laptop CLANG64
$ ./a.exe
sum: 9
```

question 2

```
int ECSumList(const set<int> &listNums)
{
    // your code goes here ...
    int sum = 0;
    for (set<int>::const_iterator it = listNums.begin(); it != listNums.end(); ++it) {
        sum += *it;
    }
    return sum;
}
```

```
karin@Karinas_Laptop CLANG64
$ clang++ ECSumListSet.cpp
```

```
karin@Karinas_Laptop CLANG64
$ ./a.exe
sum: 9
```

```
int ECSumList(const set<int> &listNums)
{
    // your code goes here ...
    int sum = 0;
    for (int num : listNums) {
        sum += num;
    }
    return sum;
}
```

```
karin@Karinas_Laptop CLANG64
$ clang++ ECSumListSet.cpp
```

```
karin@Karinas_Laptop CLANG64
$ ./a.exe
sum: 9
```

question 4

```
void ECCountDistinct(const vector<int> &listNums, map<int,int> &mapDistinct)
{
    // your code goes here
    for (int num : listNums) {
        mapDistinct[num]++;
    }
}
```

```
karin@Karinas_Laptop CLANG64
$ clang++ ECDistinct.cpp
```

```
karin@Karinas_Laptop CLANG64
$ ./a.exe
elements in mapDist:
key:1 value:2
key:2 value:1
key:3 value:4
key:5 value:2
```

question 6

```
template <class T>
bool ECommonNumber(const vector<T> &listA, const vector<T> &listB, T &val)
{
    // your code here
    for (const T &a : listA) {
        if (find(listB.begin(), listB.end(), a) != listB.end()){
            val = a;
            return true;
        }
    }
    return false;
}
```

```
karin@Karin's_Laptop CLANG64 /c/Users/
$ clang++ ECommonNumberGen.cpp

karin@Karin's_Laptop CLANG64 /c/Users/
$ ./a.exe
Test passed: equal: 1 1
Test passed: equal: 3 3
Test passed: equal: 1 1
Test passed: equal: 3150 3150
```